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Exercise 1.2

Q1. Find the values of x and y in each of the following:

(i) $x + iy + 2 - 3i = i(5 - i)(3 + 4i)$

Sol. $x + iy + 2 - 3i = i[5(3 + 4i) - i(3 + 4i)]$

$$x + iy + 2 - 3i = i[15 + 20i - 3i - 4i^2]$$

$$x + iy + 2 - 3i = i[15 + 20i - 3i - 4(-1)]$$

$$x + iy + 2 - 3i = i[15 + 20i - 3i + 4]$$

$$x + iy + 2 - 3i = i(19 + 17i)$$

$$x + iy + 2 - 3i = 19i + 17i^2$$

$$x + iy + 2 - 3i = 19i + 17(-1)$$

$$x + iy + 2 - 3i = 19i - 17$$

$$x + iy = 19i - 17 - 2 + 3i$$

$$x + iy = -19 + 22i$$

By comparing the real and imaginary parts

$$x = -19 \text{ and } y = 22$$

$$(ii) (x + iy)(1 - i) = (2 - 3i)(-5 + 5i)(-i\frac{3}{5})$$

$$\text{Sol. } (x + iy)(1 - i) = [2(-5 + 5i) - 3i(-5 + 5i)](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = [-10 + 10i + 15i - 15i^2](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = [-10 + 10i + 15i - 15(-1)](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = [-10 + 10i + 15i - 15](-i\frac{3}{5})$$

$$(x + iy)(1 - i) = (5 + 25i)(-i\frac{3}{5})$$

$$(x + iy)(1 - i) = -3i - 15i^2$$

$$(x + iy)(1 - i) = -3i - 15(-1)$$

$$(x + iy)(1 - i) = -3i + 15$$

$$x + iy = \frac{15 - 3i}{1 - i}$$

$$x + iy = \frac{15 - 3i}{1 - i} \times \frac{1 + i}{1 + i}$$

$$x + iy = \frac{15 + 15i - 3i - 3i^2}{1^2 - i^2}$$

$$x + iy = \frac{15 + 15i - 3i - 3(-1)}{1 - (-1)}$$

$$x + iy = \frac{15 + 15i - 3i + 3}{1 + 1}$$

$$x + iy = \frac{18 + 12i}{2}$$

$$x + iy = \frac{18}{2} + \frac{12}{2}i$$

$$x + iy = 9 + 6i$$

By comparing the real and imaginary parts

$$x = 9 \text{ and } y = 6$$

$$(iii) \frac{x}{2+i} + \frac{y}{3-i} = 4 + 5i$$

$$\text{Sol. } \frac{x(3-i) + y(2+i)}{(2+i)(3-i)} = 4 + 5i$$

$$\frac{3x - xi + 2y + yi}{6 - 2i + 3i - i^2} = 4 + 5i$$

$$\frac{3x + 2y - xi + yi}{6 - 2i + 3i - (-1)} = 4 + 5i$$

$$\frac{3x + 2y - xi + yi}{6 - 2i + 3i + 1} = 4 + 5i$$

$$\frac{3x + 2y - (x - y)i}{7 + i} = 4 + 5i$$

$$3x + 2y - (x - y)i = (4 + 5i)(7 + i)$$

$$3x + 2y - (x - y)i = 28 + 4i + 35i + 5i^2$$

$$3x + 2y - (x - y)i = 28 + 4i + 35i + 5(-1)$$

$$3x + 2y - (x - y)i = 28 + 4i + 35i - 5$$

$$3x + 2y - (x - y)i = 23 + 39i$$

By comparing the real and imaginary parts

$$3x + 2y = 23 \quad \text{(i)}$$

$$-(x - y) = 39 \quad \text{(ii)}$$

From (ii)

$$-(x - y) = 39$$

$$-x + y = 39$$

$$y = 39 + x \quad \text{(iii)}$$

Put in (i)

$$3x + 2y = 23$$

$$3x + 2(39 + x) = 23$$

$$3x + 78 + 2x = 23$$

$$5x = 23 - 78$$

$$5x = -55$$

$$x = -\frac{55}{5}$$

$$x = -11$$

Put in (iii)

$$y = 39 + x$$

$$y = 39 - 11$$

$$y = 28$$

02. If $z_1 = -13 + 24i$ and $z_2 = x + yi$, find the values of x and y such that $z_1 - z_2 = -27 + 15i$

$$\text{Sol. } (-13 + 24i) - (x + yi) = -27 + 15i$$

$$-13 + 24i - x - yi = -27 + 15i$$

$$-13 + 24i + 27 - 15i = x + yi$$

$$14 + 9i = x + yi$$

By comparing the real and imaginary parts

$$x = 14 \text{ and } y = 9$$

Q3 Find the real value of x and y if:

(i) $(x + iy)^2 = 25 + 60i$

Sol. $\sqrt{(x + iy)^2} = \sqrt{25 + 60i}$

$$x + iy = \sqrt{25 + 60i}$$

By applying the square root formula for complex numbers

$$x + iy = \pm \left(\sqrt{\frac{\sqrt{25^2 + 60^2} + 25}{2}} + \frac{i(60)}{|60|} \sqrt{\frac{\sqrt{25^2 + 60^2} - 25}{2}} \right)$$

$$\therefore \sqrt{x + yi} = \pm \left(\sqrt{\frac{\sqrt{x^2 + y^2} + x}{2}} + \frac{iy}{|y|} \sqrt{\frac{\sqrt{x^2 + y^2} - x}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{\sqrt{625 + 3600} + 25}{2}} + \frac{60i}{60} \sqrt{\frac{\sqrt{625 + 3600} - 25}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{\sqrt{4225} + 25}{2}} + i \sqrt{\frac{\sqrt{4225} - 25}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{65 + 25}{2}} + i \sqrt{\frac{65 - 25}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{90}{2}} + i \sqrt{\frac{40}{2}} \right)$$

$$x + iy = \pm (\sqrt{45} + i\sqrt{20})$$

$$x + iy = \pm (3\sqrt{5} + i2\sqrt{5})$$

Either

$$x + iy = +(3\sqrt{5} + i2\sqrt{5}) \text{ or } x + iy = -(3\sqrt{5} + i2\sqrt{5})$$

$$x + iy = 3\sqrt{5} + i2\sqrt{5} \text{ or } x + iy = -3\sqrt{5} - i2\sqrt{5}$$

By comparing the real and imaginary parts

$$x = 3\sqrt{5} \text{ and } y = 2\sqrt{5} \text{ or } x = -3\sqrt{5} \text{ and } y = -2\sqrt{5}$$

(ii) $(x + iy)^2 = 64 + 48i$

Sol. $\sqrt{(x + iy)^2} = \sqrt{64 + 48i}$

$$x + iy = \sqrt{64 + 48i}$$

By applying the square root formula for complex numbers

$$\therefore \sqrt{x + yi} = \pm \left(\sqrt{\frac{\sqrt{x^2 + y^2} + x}{2}} + \frac{iy}{|y|} \sqrt{\frac{\sqrt{x^2 + y^2} - x}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{\sqrt{64^2 + 48^2} + 64}{2}} + \frac{i(48)}{|48|} \sqrt{\frac{\sqrt{64^2 + 48^2} - 64}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{\sqrt{4096 + 2304} + 64}{2}} + \frac{48i}{48} \sqrt{\frac{\sqrt{4096 + 2304} - 64}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{\sqrt{6400} + 64}{2}} + i \sqrt{\frac{\sqrt{6400} - 64}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{80 + 64}{2}} + i \sqrt{\frac{80 - 64}{2}} \right)$$

$$x + iy = \pm \left(\sqrt{\frac{144}{2}} + i \sqrt{\frac{16}{2}} \right)$$

$$x + iy = \pm (\sqrt{72} + i\sqrt{8})$$

$$x + iy = \pm (6\sqrt{2} + i2\sqrt{2})$$

Either

$$x + iy = +(6\sqrt{2} + i2\sqrt{2}) \text{ or } x + iy = -(6\sqrt{2} + i2\sqrt{2})$$

$$x + iy = 6\sqrt{2} + i2\sqrt{2} \text{ or } x + iy = -6\sqrt{2} - i2\sqrt{2}$$

By comparing the real and imaginary parts

$$x = 6\sqrt{2} \text{ and } y = 2\sqrt{2} \text{ or } x = -6\sqrt{2} \text{ and } y = -2\sqrt{2}$$

(iii) $(x + iy)^2 = \frac{2i - 3}{3 + i}$

Sol. $\sqrt{(x + iy)^2} = \sqrt{\frac{2i - 3}{3 + i}}$

$$\begin{aligned} x + iy &= \sqrt{\frac{2i - 3}{3 + i} \times \frac{3 - i}{3 - i}} = \sqrt{\frac{(2i - 3)(3 - i)}{(3)^2 + i^2}} \\ &= \sqrt{\frac{(2i - 3)(3 - i)}{9 + 1}} = \sqrt{\frac{2i(3 - i) - 3(3 - i)}{10}} \\ &= \sqrt{\frac{6i - 2i^2 - 9 + 3i}{10}} = \sqrt{\frac{6i + 2 - 9 + 3i}{10}} \\ &= \sqrt{\frac{-7 + 9i}{10}} = \sqrt{\frac{-7}{10} + \frac{9}{10}i} \end{aligned}$$

By applying the square root formula for complex numbers

$$\begin{aligned} x + iy &= \pm \left(\sqrt{\frac{\left(\frac{-7}{10}\right)^2 + \left(\frac{9}{10}\right)^2 - \frac{7}{10}}{2}} + \frac{i \frac{9}{10}}{\frac{9}{10}} \sqrt{\frac{\left(\frac{-7}{10}\right)^2 + \left(\frac{9}{10}\right)^2 + \frac{7}{10}}{2}} \right) \\ &= \pm \left(\sqrt{\frac{\frac{49}{100} + \frac{81}{100} - \frac{7}{10}}{2}} + i \sqrt{\frac{\frac{49}{100} + \frac{81}{100} + \frac{7}{10}}{2}} \right) \end{aligned}$$

$$\begin{aligned}
&= \pm \left(\sqrt{\frac{\sqrt{130}-7}{2}} + i \sqrt{\frac{\sqrt{130}+7}{2}} \right) \\
&= \pm \left(\sqrt{\frac{\sqrt{130}-7}{10 \times 10}} + i \sqrt{\frac{\sqrt{130}+7}{10 \times 10}} \right) \\
&= \pm \left(\sqrt{\frac{\sqrt{130}-7}{10 \times 2}} + i \sqrt{\frac{\sqrt{130}+7}{10 \times 2}} \right) \\
&= \pm \left(\sqrt{\frac{\sqrt{130}-7}{20}} + i \sqrt{\frac{\sqrt{130}+7}{20}} \right)
\end{aligned}$$

Either

$$x+iy = + \left(\sqrt{\frac{\sqrt{130}-7}{20}} + i \sqrt{\frac{\sqrt{130}+7}{20}} \right)$$

or $x+iy = - \left(\sqrt{\frac{\sqrt{130}-7}{20}} + i \sqrt{\frac{\sqrt{130}+7}{20}} \right)$

By comparing the real and imaginary parts

$$x = \sqrt{\frac{\sqrt{130}-7}{20}} \quad \text{and} \quad y = \sqrt{\frac{\sqrt{130}+7}{20}}$$

or $x = -\sqrt{\frac{\sqrt{130}-7}{20}} \quad \text{and} \quad y = -\sqrt{\frac{\sqrt{130}+7}{20}}$

Q4. If $z_1 = 2 + 3i$ and $z_2 = 1 - \alpha$, find the value of α such that $\text{Im}(z_1 z_2) = 7$.

Sol. For $z_1 z_2$

$$z_1 z_2 = (2 + 3i)(1 - \alpha)$$

$$z_1 z_2 = 2 - 2\alpha + 3i - 3\alpha i$$

$$z_1 z_2 = (2 - 2\alpha) + (3 - 3\alpha)i$$

$$\text{Re}(z_1 z_2) = 2 - 2\alpha \quad \text{and} \quad \text{Im}(z_1 z_2) = 3 - 3\alpha$$

$$\text{Given } \text{Im}(z_1 z_2) = 7$$

$$\text{Put the value of } \text{Im}(z_1 z_2)$$

$$3 - 3\alpha = 7$$

$$-3\alpha = 7 - 3$$

$$-3\alpha = 4$$

$$\alpha = \frac{4}{-3}$$

$$\alpha = -\frac{4}{3}$$

Q5. If $z_1 = x + yi$ and $z_2 = a + bi$, find x, y, a and b such that $z_1 + z_2 = 10 + 4i$ and $z_1 - z_2 = 6 + 2i$.

Sol. Given that $z_1 + z_2 = 10 + 4i$

$$x + yi + a + bi = 10 + 4i$$

$$x + a + yi + bi = 10 + 4i$$

$$(x + a) + (y + b)i = 10 + 4i$$

By comparing the real and imaginary parts

$$x + a = 10 \quad \text{(i)}$$

$$y + b = 4 \quad \text{(ii)}$$

Also

$$z_1 - z_2 = 6 + 2i$$

$$x + yi - (a + bi) = 6 + 2i$$

$$x + yi - a - bi = 6 + 2i$$

$$x - a + yi - bi = 6 + 2i$$

$$(x - a) + (y - b)i = 6 + 2i$$

By comparing the real and imaginary parts

$$x - a = 6 \quad \text{(i)}$$

$$y - b = 2 \quad \text{(ii)}$$

Add (i) and (ii)

$$x + a + x - a = 10 + 6$$

$$2x = 16 \Rightarrow x = \frac{16}{2} \Rightarrow x = 8$$

Put in (i)

$$x + a = 10$$

$$8 + a = 10$$

$$a = 10 - 8$$

$$a = 2$$

Add (ii) and (iv)

$$y + b + y - b = 4 + 2$$

$$2y = 6 \Rightarrow y = \frac{6}{2} \Rightarrow y = 3$$

Put in (ii)

$$y + b = 4$$

$$3 + b = 4$$

$$b = 4 - 3$$

$$b = 1$$

Q6. Show that $\forall z_1, z_2 \in \mathbb{C}, \overline{z_1 z_2} = \overline{z_1} \overline{z_2}$

Sol. Assume that $z_1 = a_1 + b_1 i$ and $z_2 = a_2 + b_2 i$

$$\text{L.H.S} = \overline{z_1 z_2}$$

$$= \overline{(a_1 + b_1 i)(a_2 + b_2 i)}$$

$$= \overline{a_1 a_2 + a_1 b_2 i + a_2 b_1 i + b_1 b_2 i^2}$$

$$= \overline{a_1 a_2 + a_1 b_2 i + a_2 b_1 i + b_1 b_2 (-1)}$$

$$= \overline{a_1 a_2 + a_1 b_2 i + a_2 b_1 i - b_1 b_2}$$

$$= \overline{a_1 a_2 - b_1 b_2 + a_1 b_2 i + a_2 b_1 i}$$

$$= (a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1)i$$

$$= (a_1a_2 - b_1b_2) - (a_1b_2 + a_2b_1)i \quad \text{--- (i)}$$

$$\text{R.H.S} = z_1 z_2$$

$$= (\overline{a_1 + b_1 i})(\overline{a_2 + b_2 i})$$

$$= a_1a_2 - a_1b_2i - a_2b_1i + b_1b_2i^2$$

$$= a_1a_2 - a_1b_2i - a_2b_1i + b_1b_2(-1)$$

$$= a_1a_2 - a_1b_2i - a_2b_1i - b_1b_2$$

$$= a_1a_2 - b_1b_2 - a_1b_2i - a_2b_1i$$

$$= (a_1a_2 - b_1b_2) - (a_1b_2 + a_2b_1)i \quad \text{--- (ii)}$$

From (i) and (ii)

L.H.S = R.H.S

Q7 Find the square root of the following complex numbers:

(i) $-7 - 24i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x + yi} = \pm \left(\sqrt{\frac{\sqrt{x^2 + y^2} + x}{2}} + \frac{iy}{|y|} \sqrt{\frac{\sqrt{x^2 + y^2} - x}{2}} \right)$$

$$\sqrt{-7 - 24i} =$$

$$\pm \left(\sqrt{\frac{\sqrt{(-7)^2 + (-24)^2} + (-7)}{2}} + \frac{i(-24)}{|-24|} \sqrt{\frac{\sqrt{(-7)^2 + (-24)^2} - (-7)}{2}} \right)$$

$$\sqrt{-7 - 24i} = \pm \left(\sqrt{\frac{\sqrt{49 + 576} - 7}{2}} - \frac{24i}{24} \sqrt{\frac{\sqrt{49 + 576} + 7}{2}} \right)$$

$$\sqrt{-7 - 24i} = \pm \left(\sqrt{\frac{\sqrt{625} - 7}{2}} - i \sqrt{\frac{\sqrt{625} + 7}{2}} \right)$$

$$\sqrt{-7 - 24i} = \pm \left(\sqrt{\frac{25 - 7}{2}} - i \sqrt{\frac{25 + 7}{2}} \right)$$

$$\sqrt{-7 - 24i} = \pm \left(\sqrt{\frac{18}{2}} - i \sqrt{\frac{32}{2}} \right)$$

$$\sqrt{-7 - 24i} = \pm (\sqrt{9} - i\sqrt{16})$$

$$\sqrt{-7 - 24i} = \pm (3 - 4i)$$

Either

$$\sqrt{-7 - 24i} = +(3 - 4i) \quad \text{or} \quad \sqrt{-7 - 24i} = -(3 - 4i)$$

$$\sqrt{-7 - 24i} = 3 - 4i \quad \text{or} \quad \sqrt{-7 - 24i} = -3 + 4i$$

(ii) $8 - 6i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x + yi} = \pm \left(\sqrt{\frac{\sqrt{x^2 + y^2} + x}{2}} + \frac{iy}{|y|} \sqrt{\frac{\sqrt{x^2 + y^2} - x}{2}} \right)$$

$$\sqrt{8 - 6i} =$$

$$\pm \left(\sqrt{\frac{\sqrt{(8)^2 + (-6)^2} + (8)}{2}} + \frac{i(-6)}{|-6|} \sqrt{\frac{\sqrt{(8)^2 + (-6)^2} - (8)}{2}} \right)$$

$$\sqrt{8 - 6i} = \pm \left(\sqrt{\frac{\sqrt{64 + 36} + 8}{2}} - \frac{6i}{6} \sqrt{\frac{\sqrt{64 + 36} - 8}{2}} \right)$$

$$\sqrt{8 - 6i} = \pm \left(\sqrt{\frac{\sqrt{100} + 8}{2}} - i \sqrt{\frac{\sqrt{100} - 8}{2}} \right)$$

$$\sqrt{8 - 6i} = \pm \left(\sqrt{\frac{10 + 8}{2}} - i \sqrt{\frac{10 - 8}{2}} \right)$$

$$\sqrt{8 - 6i} = \pm \left(\sqrt{\frac{18}{2}} - i \sqrt{\frac{2}{2}} \right)$$

$$\sqrt{8 - 6i} = \pm (\sqrt{9} - i\sqrt{1})$$

$$\sqrt{8 - 6i} = \pm (3 - i)$$

Either

$$\sqrt{8 - 6i} = +(3 - i) \quad \text{or} \quad \sqrt{8 - 6i} = -(3 - i)$$

$$\sqrt{8 - 6i} = 3 - i \quad \text{or} \quad \sqrt{8 - 6i} = -3 + i$$

(iii) $-15 - 36i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x + yi} = \pm \left(\sqrt{\frac{\sqrt{x^2 + y^2} + x}{2}} + \frac{iy}{|y|} \sqrt{\frac{\sqrt{x^2 + y^2} - x}{2}} \right)$$

$$\sqrt{-15 - 36i} = \pm$$

$$\pm \left(\sqrt{\frac{\sqrt{(-15)^2 + (-36)^2} + (-15)}{2}} + \frac{i(-36)}{|-36|} \sqrt{\frac{\sqrt{(-15)^2 + (-36)^2} - (-15)}{2}} \right)$$

$$\sqrt{-15 - 36i} = \pm \left(\sqrt{\frac{\sqrt{225 + 1296} - 15}{2}} - \frac{36i}{36} \sqrt{\frac{\sqrt{225 + 1296} + 15}{2}} \right)$$

$$\sqrt{-15 - 36i} = \pm \left(\sqrt{\frac{\sqrt{1521} - 15}{2}} - i \sqrt{\frac{\sqrt{1521} + 15}{2}} \right)$$

$$\sqrt{-15 - 36i} = \pm \left(\sqrt{\frac{39 - 15}{2}} - i \sqrt{\frac{39 + 15}{2}} \right)$$

$$\sqrt{-15 - 36i} = \pm \left(\sqrt{\frac{24}{2}} - i \sqrt{\frac{54}{2}} \right)$$

$$\sqrt{-15 - 36i} = \pm (\sqrt{12} - i\sqrt{27})$$

$$\sqrt{-15 - 36i} = \pm (2\sqrt{3} - 3\sqrt{3}i)$$

Either

$$\sqrt{-15-36i} = +(2\sqrt{3}-3\sqrt{3}i) \text{ or } \sqrt{-15-36i} = -(2\sqrt{3}-3\sqrt{3}i)$$

$$\sqrt{-15-36i} = 2\sqrt{3}-3\sqrt{3}i \text{ or } \sqrt{-15-36i} = -2\sqrt{3}+3\sqrt{3}i$$

(iv) $119 + 120i$

Sol. By using the square root formula for complex numbers

$$\sqrt{x+yi} = \pm \left(\sqrt{\frac{x^2+y^2+x}{2}} + \frac{iy}{|y|} \sqrt{\frac{x^2+y^2-x}{2}} \right)$$

$$\sqrt{119+120i} = \pm$$

$$\left(\sqrt{\frac{(119)^2+(120)^2+(119)}{2}} + \frac{i(120)}{|120|} \sqrt{\frac{(119)^2+(120)^2-(119)}{2}} \right)$$

$$\sqrt{119+120i} = \pm$$

$$\left(\sqrt{\frac{\sqrt{14161+14400}+119}{2}} + \frac{120i}{120} \sqrt{\frac{\sqrt{14161+14400}-119}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{\sqrt{28561}+119}{2}} + i \sqrt{\frac{\sqrt{28561}-119}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{169+119}{2}} + i \sqrt{\frac{169-119}{2}} \right)$$

$$\sqrt{119+120i} = \pm \left(\sqrt{\frac{288}{2}} + i \sqrt{\frac{50}{2}} \right)$$

$$\sqrt{119+120i} = \pm (\sqrt{144} + i\sqrt{25})$$

$$\sqrt{119+120i} = \pm (12+5i)$$

Either

$$\sqrt{119+120i} = +(12+5i) \text{ or } \sqrt{119+120i} = -(12+5i)$$

$$\sqrt{119+120i} = 12+5i \text{ or } \sqrt{119+120i} = -12-5i$$

Q8. Find the square root of $13-20\sqrt{3}i$ and represent them on an Argand diagram.

Sol. By using the square root formula for complex numbers

$$\sqrt{x+yi} = \pm \left(\sqrt{\frac{x^2+y^2+x}{2}} + \frac{iy}{|y|} \sqrt{\frac{x^2+y^2-x}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm$$

$$\left(\sqrt{\frac{(13)^2+(-20\sqrt{3})^2+(13)}{2}} + \frac{i(-20\sqrt{3})}{|-20\sqrt{3}|} \sqrt{\frac{(13)^2+(-20\sqrt{3})^2-(13)}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm$$

$$\left(\sqrt{\frac{\sqrt{169+1200}+13}{2}} - \frac{20\sqrt{3}i}{20\sqrt{3}} \sqrt{\frac{\sqrt{169+1200}-13}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{\sqrt{1369}+13}{2}} - i \sqrt{\frac{\sqrt{1369}-13}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{37+13}{2}} - i \sqrt{\frac{37-13}{2}} \right)$$

$$\sqrt{13-20\sqrt{3}i} = \pm \left(\sqrt{\frac{50}{2}} - i \sqrt{\frac{24}{2}} \right)$$

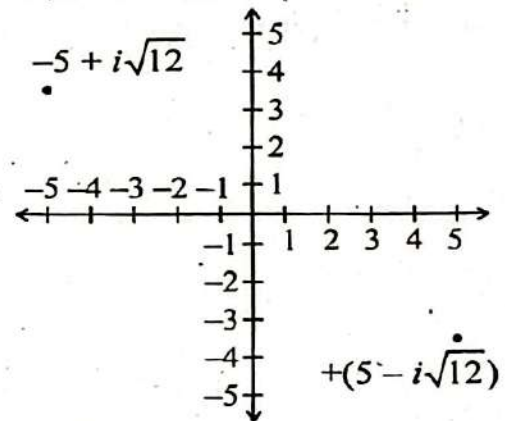
$$\sqrt{13-20\sqrt{3}i} = \pm (\sqrt{25} - i\sqrt{12})$$

$$\sqrt{13-20\sqrt{3}i} = \pm (5-2\sqrt{3}i)$$

Either

$$\sqrt{13-20\sqrt{3}i} = +(5-2\sqrt{3}i) \text{ or } \sqrt{13-20\sqrt{3}i} = -(5-2\sqrt{3}i)$$

$$\sqrt{13-20\sqrt{3}i} = 5-2\sqrt{3}i \text{ or } \sqrt{13-20\sqrt{3}i} = -5+2\sqrt{3}i$$



Q9. Find the value of x and y if

$$(-7+i)(x+iy) + (-1-5i) = i(11-i)$$

$$\text{Sol. } (-7+i)(x+iy) - 1-5i = 11i-i^2$$

$$(-7+i)(x+iy) - 1-5i = 11i - (-1)$$

$$(-7+i)(x+iy) - 1-5i = 11i + 1$$

$$(-7+i)(x+iy) = 11i + 1 + 1 + 5i$$

$$(-7+i)(x+iy) = 2 + 16i$$

$$x+iy = \frac{2+16i}{-7+i}$$

$$x+iy = \frac{2+16i}{-7+i} \times \frac{-7-i}{-7-i}$$

$$x+iy = \frac{-14-2i-112i-16i^2}{(-7)^2-i^2}$$

$$x+iy = \frac{-14-2i-112i-16(-1)}{49-(-1)}$$

$$x+iy = \frac{-14-2i-112i+16}{49+1}$$

$$x + iy = \frac{2 - 114i}{50}$$

$$x + iy = \frac{2}{50} - \frac{114}{50}i$$

$$x + iy = \frac{1}{25} - \frac{57}{25}i$$

By comparing the real and imaginary parts

$$x = \frac{1}{25} \text{ and } y = -\frac{57}{25}$$

Q10. Find the value of x and y if

$$(5 - 2i)(x + iy) + 3 = i(11 - i) - 4i$$

Sol. $(5 - 2i)(x + iy) + 3 = 11i - i^2 - 4i$

$$(5 - 2i)(x + iy) + 3 = 11i - (-1) - 4i$$

$$(5 - 2i)(x + iy) = 11i + 1 - 4i - 3$$

$$(5 - 2i)(x + iy) = -2 + 7i$$

$$x + iy = \frac{-2 + 7i}{5 - 2i}$$

$$x + iy = \frac{-2 + 7i}{5 - 2i} \times \frac{5 + 2i}{5 + 2i}$$

$$x + iy = \frac{-10 - 4i + 35i + 14i^2}{5^2 - (2i)^2}$$

$$x + iy = \frac{-10 - 4i + 35i + 14i^2}{25 - 4i^2}$$

$$x + iy = \frac{-10 - 4i + 35i + 14(-1)}{25 - 4(-1)}$$

$$x + iy = \frac{-10 - 4i + 35i - 14}{25 + 4}$$

$$x + iy = \frac{-24 + 31i}{29}$$

$$x + iy = -\frac{24}{29} + \frac{31}{29}i$$

By comparing the real and imaginary parts

$$x = -\frac{24}{29} \text{ and } y = \frac{31}{29}$$

Q11. Find the values of u and v if $\frac{u-2}{2+i} + \frac{v-3}{2-i} = 4i$

Sol. $\frac{u-2}{2+i} + \frac{v-3}{2-i} = 4i$

$$\frac{(u-2)(2-i) + (v-3)(2+i)}{(2+i)(2-i)} = 4i$$

$$\frac{2u - ui - 4 + 2i + 2v + vi - 6 - 3i}{2^2 - i^2} = 4i$$

$$\frac{2u - ui + 2v + vi - 10 - i}{4 - (-1)} = 4i$$

$$\frac{2u + 2v - 10 - ui + vi - i}{4 + 1} = 4i$$

$$\frac{(2u + 2v - 10) + i(-u + v - 1)}{5} = 4i$$

$$\frac{2u + 2v - 10}{5} + \frac{-u + v - 1}{5}i = 0 + 4i$$

By comparing both sides

$$\frac{2u + 2v - 10}{5} = 0 \quad \dots(i)$$

$$\frac{-u + v - 1}{5} = 4 \quad \dots(ii)$$

From (ii)

$$-u + v - 1 = 4 \times 5$$

$$-u + v - 1 = 20$$

$$v = 20 + 1 + u$$

$$v = 21 + u \quad (iii)$$

Put in (i)

$$2u + 2(21 + u) - 10 = 0 \times 5$$

$$2u + 42 + 2u - 10 = 0$$

$$4u + 32 = 0$$

$$4u = -32$$

$$u = -\frac{32}{4}$$

$$u = -8$$

Put in (iii)

$$v = 21 + (-8)$$

$$v = 21 - 8$$

$$v = 13$$

Q12. If $z_1 = 4 + 5i$ and $z_2 = \alpha - 2i$, find the value of α such that $\text{Re}(z_1 z_2) = 20$.

Sol. For $z_1 z_2$

$$z_1 z_2 = (4 + 5i)(\alpha - 2i)$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i - 10i^2$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i - 10(-1)$$

$$z_1 z_2 = 4\alpha - 8i + 5\alpha i + 10$$

$$z_1 z_2 = 4\alpha + 10 - 8i + 5\alpha i$$

$$z_1 z_2 = (4\alpha + 10) - (8 - 5\alpha)i$$

$$\text{Re}(z_1 z_2) = 4\alpha + 10 \text{ and } \text{Im}(z_1 z_2) = -(8 - 5\alpha)$$

$$\text{Given } \text{Re}(z_1 z_2) = 20$$

Put the value of $\text{Re}(z_1 z_2)$

$$4\alpha + 10 = 20$$

$$4\alpha = 20 - 10$$

$$4\alpha = 10$$

$$\alpha = \frac{10}{4} \Rightarrow \alpha = \frac{5}{2}$$